DAY04-TCPIP-OSI

OSI/TCP-IP THEORY (10 marks)

1. What is a network protocol?
   Set of rules followed when communicating over the internet
   Eg: HTTP, HTTPS, DNS, UDP, IP, TCP, Ethernet

2. What is the OSI Model?
   OSI model also known as open-source interconnection model
   It is a conceptual framework that breaks down network communication into seven distinct layers

3. Name all seven OSI layers in correct order.
   - Physical
   - Data link
   - Transport
   - Network
   - Session
   - Presentable
   - Application

4. What is the purpose of the Application Layer?
   - It allows or lets network communication meet applications on how to format, send, interpret data
   - Through browsing web page by HTTP/HTTPS.
   - Transferring files through FTP/TFTP.
   - Sending/ Receiving email through SMTP/POP3/IMAP

5. What is the purpose of the Transport Layer?
   - It does the end-to-end communication between application processes also called process to process
   - It uses port number to identify process on each host.
     -UDP- User Datagram Protocol- simple and efficient
     -TCP-Transmission Control Protocol-more robust features beyond message addressing.

6. What is the purpose of the Network Layer?
   - It does the end-to-end delivery between hosts across multiple networks
   - Uses IP address to identify host in network
   - Categorized as two where IP into IPv4, IPv6 also ICMP Internet control message protocol
7. What is the purpose of the Data Link Layer?
   The Data Link Layer ensures that data is properly packaged and delivered from one device to another on the same network.

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8. What is the purpose of the Physical Layer?
    - Sends and receives bit as electrical, optical or radio over medium
    - Cables, connectors, signal levels and link speeds are mostly done by them
    - Eg: Copper UTP cables, Fiber-optic cables, NIC, Wi-Fi, Radio and antennas

9. What is the TCP/IP Model?
   The TCP/IP Model is a networking framework used to describe how data is transmitted between devices over a network, especially the Internet.
   It has 4 layers Application Layer, Transport Layer, Internet layer, Network Access layer

10. Explain the difference between OSI and TCP/IP models.

OSI model	TCP/IP model
Has 7 layers	Has 4 layers
Has separate **Session** and **Presentation** layers	These are included in the **Application Layer**
Has a separate **Physical** and **Data Link** layer	These are combined into the **Network Access Layer**
More detailed	Simpler and more practical

11. What is encapsulation?

As data moves DOWN the stack, each layer ADDS its header:

Application Data → +Transport Header (TCP/UDP) → +IP Header → +Ethernet Header → Bits

12. What is decapsulation?

On the receiving end, headers are processed and removed moving UP the stack:

Bits → Ethernet Frame → IP Packet → TCP/UDP Segment → Application Data

13. Which layer uses IP addresses?

- Internet Layer

14. Which layer uses MAC addresses?

- Local Network Layer

15. Which layer uses TCP and UDP?

- Transport Layer

16. Which layer includes HTTP and HTTPS?

- Application Layer

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17. Why should cybersecurity professionals understand networking layers?

Cybersecurity professionals need to understand networking layers because attacks and security controls can occur at different layers of a network.

18. Explain how network data travels from a browser to a server.

The browser first uses DNS to find the server's IP address, then sends an HTTP/HTTPS request through the network using TCP/IP. Routers forward the data to the server, which processes the request and sends a response back to the browser.

Browser → DNS → TCP/TLS → HTTP Request → IP Packets → Router → Server → Response → Browser

PRACTICAL (4 marks)

19. Your resolved IP address for google.com (nslookup).

Server: dns9.quad9.net

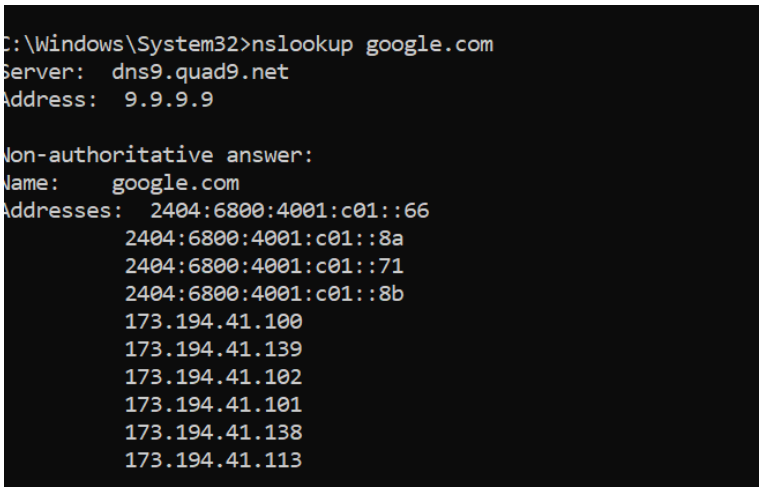
Name google.com

20. Number of traceroute hops to google.com

- 173.194.41.100
- 173.194.41.139
- 173.194.41.102
- 173.194.41.101
- 173.194.41.138
- 173.194.41.113

SCREENSHOTS (3 marks)

Attach screenshots of: ping google.com, tracert google.com, nslookup google.com



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```
C:\Windows\System32>ping google.com

Pinging google.com [173.194.41.101] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 173.194.41.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Windows\System32>tracert google.com

Tracing route to google.com [173.194.41.100]
over a maximum of 30 hops:

  1     2 ms     3 ms     2 ms  10.101.128.1
  2    20 ms     2 ms     4 ms  192.168.200.1
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *     ^C
C:\Windows\System32>
```